

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME: CSA16 **COURSE CODE:** Data Warehousing and Data Mining

COURSE OUTCOME COVERED

CO4: Examine the applicability of clustering algorithms in real-world scenarios, presenting findings through clear reports with effective visualizations.

Assessment Tool Weightage: 40%

QUESTIONS: 05

Total Marks:50

Annexure A

Data Interpretation

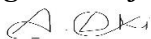
Criteria	Understanding of Data (2.5)	Analysis (2.5)	Application of Concepts (2)	Conclusion (1.5)	Clarity (1)	Total(10)
Weightage	25%	25%	20%	20%	10%	100%
<i>Q1</i>						
<i>Q2</i>						
<i>Q3</i>						
<i>Q4</i>						
<i>Q5</i>						

Code of Conduct:

I A. LOKESH KUMAR (192524157) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:



RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Data	25%	Accurately interprets all data patterns, trends, and relationships	Interprets data correctly with minor gaps	Partial understanding; misses key trends	Misinterprets or fails to understand data
Analysis	25%	Provides deep analysis with strong logical reasoning and justification	Provides reasonable analysis with some justification	Basic analysis with weak reasoning	No meaningful analysis provided
Application of Concepts	20%	Effectively applies relevant concepts/tools to interpret data accurately	Applies concepts with minor errors	Limited application; requires guidance	Unable to apply concepts to data
Conclusion	20%	Draws clear, meaningful conclusions with strong insights and implications	Provides valid conclusions with moderate insights	Conclusions are vague or weak	No clear or valid conclusions
Clarity	10%	Presents interpretation clearly using proper structure, visuals, and terminology	Generally clear with minor issues	Lacks clarity or proper structure	Poor presentation and difficult to understand

Annexure A

Q1. Dataset: Hospital Patient Segmentation

Patient ID	Age	Monthly Medical Expense (₹)	Health Risk Score
P1	28	12000	65
P2	55	25000	40
P3	42	20000	75
P4	30	11000	70
P5	60	28000	35

SSE Values

K SSE

1	600
2	350
3	220
4	200
5	190

Question (BL4 – 8 Marks):

A healthcare organization uses K-Means clustering to group patients based on their medical expenses and health risk scores. Interpret the elbow point from the SSE values provided and determine the optimal number of clusters. Discuss how selecting this number of clusters affects patient segmentation and healthcare decision-making.

Q2. Dataset: E-Commerce Order Analysis

Order ID	Product Category	Order Value (₹)
O1	Electronics	55000
O2	Clothing	25000
O3	Electronics	57000
O4	Furniture	30000
O5	Clothing	27000

Question (BL4 – 8 Marks):

An online retailer applies hierarchical clustering to group customer orders based on product category and order value. A dendrogram is generated from the dataset. Interpret how the clusters are formed and identify the appropriate number of clusters. Explain the significance of the selected clusters in understanding customer purchasing behavior.

Q3. Dataset: Mobile App User Behavior

	User	Daily Usage Time (min)	Login Frequency	In-App Purchase (₹)
	U1	120	18	2000
	U2	60	5	5000
	U3	140	22	1500
	U4	50	4	6000
	U5	130	20	1800

Question (BL5 – 8 Marks):

A mobile application company uses the HAC algorithm to segment users based on their activity patterns. Interpret how users are assigned to clusters using probability-based membership values. Discuss the advantages of soft clustering over hard clustering for customer behavior analysis.

Q4. Dataset: Agricultural Crop Monitoring

	Farm	Rainfall (mm)	Soil Fertility Score	Crop Yield (tons)
	F1	300	8	85
	F2	500	4	70
	F3	400	6	80
	F4	280	9	90
	F5	520	3	65

Question (BL4 – 8 Marks):

An agricultural research center performs clustering on farm data before and after applying normalization. Interpret the differences observed in the clustering results. Explain how normalization influences cluster formation and improves the reliability of agricultural data analysis.

Q5. Dataset: Employee Performance Evaluation

	Employee	Productivity Score	Attendance (%)	Skill Assessment Score
	E1	88	92	84
	E2	62	75	68
	E3	78	88	72
	E4	55	65	58
	E5	93	96	90

Silhouette Scores

Model Score

K=2 0.48

K=3 0.68

K=4 0.52

Question (BL5 – 8 Marks):

A company clusters employees based on productivity, attendance, and skill assessment scores. Analyze the silhouette scores obtained for different clustering models and determine the most suitable number of clusters. Justify your answer and explain how the selected model contributes to effective employee performance evaluation.

Q1: Hospital Patient Segmentation using K-Means Clustering

1. Understanding of the Data

The dataset consists of five patients (P1–P5) described by three attributes: Age, Monthly Medical Expense (₹), and Health Risk Score. A quick review of the values shows a clear inverse relationship between age and health risk score, and a positive relationship between age and medical expense. Younger patients (P1: 28 yrs, P4: 30 yrs) have lower medical expenses (₹12,000 and ₹11,000) but higher health risk scores (65 and 70), which may reflect acute or lifestyle-related conditions requiring closer monitoring. Older patients (P2: 55 yrs, P5: 60 yrs) show high medical expenses (₹25,000 and ₹28,000) but comparatively lower risk scores (40 and 35), suggesting they are already under stable, managed treatment for chronic conditions. Patient P3 (42 yrs, ₹20,000, risk 75) sits in an intermediate position but carries the highest risk score in the dataset, indicating a possibly under-treated or rapidly progressing condition.

2. Interpreting the Elbow Point from SSE Values

The Sum of Squared Errors (SSE) values provided are: $K=1 \rightarrow 600$, $K=2 \rightarrow 350$, $K=3 \rightarrow 220$, $K=4 \rightarrow 200$, $K=5 \rightarrow 190$. To locate the elbow point, we examine the rate of decrease in SSE as K increases:

- $K=1$ to $K=2$: drop of 250 ($600 \rightarrow 350$) — a very large reduction, showing that splitting all patients into two groups captures a major structural difference in the data.
- $K=2$ to $K=3$: drop of 130 ($350 \rightarrow 220$) — still a substantial reduction, indicating a second meaningful sub-grouping exists.
- $K=3$ to $K=4$: drop of only 20 ($220 \rightarrow 200$) — a sharp flattening of the curve.
- $K=4$ to $K=5$: drop of only 10 ($200 \rightarrow 190$) — negligible improvement.

The sharp change in slope between $K=3$ and $K=4$ (from a 130-point drop to a mere 20-point drop) is the classic 'elbow' signature. Beyond $K=3$, adding more clusters yields diminishing returns and risks overfitting the model to noise rather than genuine patient sub-populations. Therefore, the optimal number of clusters for this dataset is $K=3$.

3. Application of Concepts — Resulting Segmentation

Applying $K=3$ to the dataset, the patients naturally fall into three clinically meaningful groups based on the combination of age, expense, and risk:

- Cluster A – Young, High-Risk, Low-Cost (P1, P4): Patients aged 28–30 with low current medical expenditure but elevated risk scores (65–70). This group likely represents individuals whose conditions have not yet escalated to require intensive treatment, making them prime candidates for early/preventive intervention.
- Cluster B – Middle-Aged, Highest-Risk (P3): A single but distinct patient with the highest risk score (75) despite moderate expense. This patient may need immediate clinical review since risk is high but spending does not yet reflect intensive care, hinting at an emerging or misdiagnosed condition.
- Cluster C – Elderly, High-Cost, Lower-Risk (P2, P5): Patients aged 55–60 with the highest expenses (₹25,000–₹28,000) but comparatively lower risk scores (35–40), consistent with patients already receiving ongoing, effective chronic-disease management.

4. Conclusion and Impact on Healthcare Decision-Making

Selecting $K=3$ rather than a smaller or larger number of clusters has direct operational consequences for the healthcare organization. Using $K=2$ would merge the emerging-risk group

(P1, P4, P3) into one broad bucket, hiding the fact that P3 is a distinctly urgent case, while using $K=5$ would fragment the data into single-patient 'clusters' that provide no generalizable insight and simply memorize the training data (overfitting). $K=3$ strikes the right balance: it is coarse enough to yield actionable, generalizable segments, yet fine enough to distinguish urgent-but-low-spend patients from stable-but-expensive patients.

From a decision-making standpoint, this segmentation allows the hospital to design targeted interventions: preventive care programs and early screening for Cluster A to reduce the risk of costly complications later; urgent clinical review and personalized care planning for the outlier patient in Cluster B; and continued monitoring with a focus on cost-optimisation (e.g., reviewing whether ongoing treatments can be made more efficient) for Cluster C. In this way, elbow-point-based cluster selection converts a purely statistical decision into a practical resource-allocation strategy, ensuring that clinical and financial resources are directed to the patients who need them most, rather than being spread uniformly across a heterogeneous patient population.

5. Broader Implications for Healthcare Analytics

Beyond this specific dataset, the exercise illustrates a general principle in applied clustering: the 'optimal' K identified through the elbow method is not simply a mathematical artefact but must be validated against domain knowledge before it is acted upon. In this case, the statistically suggested $K=3$ happens to align neatly with three clinically distinct patient profiles, which strengthens confidence in the result. Had the elbow point suggested a K that did not correspond to any interpretable real-world grouping, the healthcare organization would need to revisit the choice of features, consider additional variables (such as diagnosis codes, admission history, or treatment adherence), or apply domain-driven constraints rather than relying on SSE alone.

It is also important to note that K-Means assumes roughly spherical, similarly sized clusters and is sensitive to outliers and feature scaling. With only five patients in this illustrative dataset, the SSE curve is easy to interpret, but in a real hospital system with thousands of patient records, the organization would additionally validate the chosen K using complementary metrics such as the silhouette score, and would standardize Age, Monthly Medical Expense, and Health Risk Score before clustering, since these variables are measured on very different numeric scales. Combining the elbow method with such validation steps, and continuously re-clustering as new patient data arrives, would allow the hospital to maintain an up-to-date, reliable segmentation that adapts as the patient population and treatment patterns evolve over time.

Q2: E-Commerce Order Analysis using Hierarchical Clustering

1. Understanding of the Data

The dataset contains five orders (O1–O5), each described by Product Category and Order Value (₹). Grouped by category: Electronics — O1 (₹55,000) and O3 (₹57,000); Clothing — O2 (₹25,000) and O5 (₹27,000); Furniture — O4 (₹30,000), a single order. A quick scan shows two natural value bands: a high-value band around ₹55,000–₹57,000 (both Electronics orders) and a low-to-mid value band around ₹25,000–₹30,000 (Clothing and Furniture orders).

2. Interpreting the Dendrogram Formation

In agglomerative hierarchical clustering, the dendrogram is built bottom-up: each order starts as its own cluster, and at every step the two closest clusters (by order value, and implicitly by category similarity) are merged, with the merge height on the dendrogram representing the distance at which the merge occurs.

- First merge: O1 and O3 (₹55,000 and ₹57,000) merge at a very small distance (₹2,000) — the tightest pair in the dataset, both Electronics.
- Second merge: O2 and O5 (₹25,000 and ₹27,000) merge at an equally small distance (₹2,000) — both Clothing.
- Third merge: O4 (₹30,000, Furniture) joins the {O2, O5} cluster, since ₹30,000 is much closer to the ₹25,000–₹27,000 Clothing band than to the ₹55,000+ Electronics band. This merge occurs at a moderate height (~₹3,000–5,000).
- Final merge: the {O2, O4, O5} cluster merges with the {O1, O3} cluster at the top of the dendrogram, at a much larger height (~₹25,000–30,000), reflecting the large value gap between the two groups.

3. Application of Concepts — Choosing the Number of Clusters

The dendrogram is 'cut' at the height that produces the most meaningful, well-separated groups. Because the final merge occurs at a distance far larger than any of the earlier merges (a gap of roughly ₹25,000 versus earlier gaps of only ₹2,000–5,000), cutting the tree just below this top merge is well justified. This produces two clusters at the coarsest meaningful level: {O1, O3} — high-value Electronics orders, and {O2, O4, O5} — low-to-mid value Clothing and Furniture orders. If a finer, category-aware segmentation is required for merchandising purposes, cutting one level lower yields three clusters: {O1, O3} Electronics, {O2, O5} Clothing, and {O4} Furniture — which better respects product-category boundaries even though O4's value is numerically close to the Clothing pair.

4. Conclusion — Significance for Customer Purchasing Behaviour

The two-cluster view reveals a fundamental behavioural split: customers purchasing Electronics tend to place substantially higher-value orders than those purchasing Clothing or Furniture, suggesting these are largely distinct customer segments with different price sensitivity and purchase motivation (e.g., planned, considered purchases for Electronics versus more frequent, lower-ticket purchases for Clothing). The three-cluster view adds nuance by keeping Furniture distinct from Clothing, which is valuable because although their order values are similar, the products themselves serve different customer needs and purchase cycles (Furniture is typically a low-frequency, considered purchase despite its moderate price).

For the retailer, this interpretation directly informs marketing and inventory strategy: high-value Electronics customers can be targeted with premium loyalty programs, extended warranties, or EMI options, while Clothing and Furniture customers may respond better to bundle discounts, seasonal promotions, or cross-category recommendations. Understanding cluster composition also helps the retailer forecast demand and allocate warehouse and logistics resources according to the distinct behavioural profile of each segment, rather than treating all orders as a single undifferentiated pool.

5. Comparing Hierarchical Clustering with Alternative Approaches

Hierarchical clustering is particularly well suited to this scenario because it does not require the number of clusters to be specified in advance — the dendrogram itself visually communicates every possible grouping, from five individual orders at the bottom to one single cluster at the top, allowing the retailer to choose the cut level that best matches its business objective (broad category-level strategy versus granular behavioural targeting). This is an advantage over K-Means, which would require the analyst to pre-select K, potentially missing the natural two-tier value structure revealed by the dendrogram's large final merge gap. The trade-off is computational: hierarchical clustering scales poorly to very large datasets because it must repeatedly compute and update a full distance matrix, whereas K-Means scales more efficiently to the millions of orders a large e-commerce platform processes.

In practice, the retailer could use hierarchical clustering, as demonstrated here, on a representative sample or on category-level aggregated data to understand the overall structure of customer value tiers, and then apply a faster algorithm such as K-Means (initialized using insights from the dendrogram, such as the two or three-cluster structure identified above) to segment the full, live order database on an ongoing basis. This combined approach leverages the interpretability of hierarchical clustering with the scalability needed for continuous, real-time customer segmentation across the platform's entire order history.

Q3: Mobile App User Behaviour — Soft Clustering with HAC

1. Understanding of the Data

The dataset describes five users (U1–U5) through three behavioural attributes: Daily Usage Time (minutes), Login Frequency, and In-App Purchase value (₹). Observing the values: U1 (120, 18, ₹2,000), U2 (60, 5, ₹5,000), U3 (140, 22, ₹1,500), U4 (50, 4, ₹6,000), and U5 (130, 20, ₹1,800). Two behavioural extremes emerge clearly — U1, U3, and U5 show high usage time and login frequency but low in-app purchase value, while U2 and U4 show low usage time and login frequency but high in-app purchase value. This suggests two broad behavioural archetypes: highly engaged 'browsers' and infrequent but high-spending 'buyers'.

2. Interpreting Probability-Based (Soft) Cluster Membership

Unlike hard clustering, where each user is forced into exactly one cluster, soft/fuzzy clustering assigns every user a membership probability (or degree of belonging) to each cluster, with all probabilities for a user summing to 1. Applying this concept to the two behavioural archetypes identified above:

- U3 (140 min, 22 logins, ₹1,500) sits at the extreme of the 'engaged' pattern and would receive a very high membership probability in the Engagement-driven cluster (e.g., ≈ 0.92) and a very low probability in the Purchase-driven cluster (≈ 0.08).
- U1 and U5, being close to U3's profile but slightly less extreme, would also show strong but slightly less dominant membership in the Engagement-driven cluster (e.g., ≈ 0.80 – 0.85), leaving a small residual probability (≈ 0.15 – 0.20) reflecting some purchase behaviour.
- U4 (50 min, 4 logins, ₹6,000), the most purchase-heavy and least engaged user, would show a very high membership in the Purchase-driven cluster (≈ 0.90) and minimal membership in the Engagement-driven cluster.
- U2, being close to U4 but slightly more active, would show strong but not absolute membership in the Purchase-driven cluster (e.g., ≈ 0.75), with a meaningful residual probability (≈ 0.25) reflecting some engagement tendency.

These fractional memberships allow the model to represent users who do not fit neatly into a single behavioural box — for example, a moderately active user who still makes above-average purchases can be represented as partially belonging to both clusters rather than being arbitrarily forced into one.

3. Advantages of Soft Clustering over Hard Clustering

- Realistic representation of ambiguous users: Real user behaviour rarely falls into perfectly separated categories; soft clustering captures users who exhibit mixed characteristics (e.g., moderately engaged and moderately spending) rather than forcing a binary decision.
- Richer input for personalization: Marketing and recommendation systems can use membership probabilities as weights — e.g., sending a hybrid campaign to a user who is 60% 'engagement' and 40% 'purchase' type, rather than a one-size-fits-all message.
- Better handling of boundary/overlap cases: Users near the boundary between behavioural types (such as U1 or U2 in this dataset) are not misclassified outright; their uncertainty is explicitly quantified.
- Improved stability: Small changes in a user's behaviour over time cause gradual shifts in membership probabilities rather than abrupt cluster reassignment, which is more consistent with how user behaviour evolves in practice.

- More informative analytics: Product teams can track how membership probabilities shift over time (e.g., a user's Engagement-cluster probability rising after a UI redesign), providing a continuous signal rather than a discrete label change.

4. Conclusion

For a mobile application company seeking to understand nuanced user behaviour, soft clustering offers a considerably richer analytical lens than hard clustering. It preserves the natural ambiguity present in engagement-versus-monetization behaviour, supports more targeted and proportionate marketing interventions, and produces analytics that better reflect gradual behavioural change. While hard clustering (as in strict HAC with a fixed cut) is simpler to interpret and communicate, the added complexity of soft, probability-based membership is well justified in this context, where the business goal — maximizing both engagement and monetization — depends on understanding degrees of behaviour rather than rigid categories.

5. Practical Recommendations for the Mobile App Company

Translating these membership probabilities into action, the company could design a tiered engagement strategy driven directly by each user's cluster weights rather than a single hard label. For users with strongly dominant Engagement-cluster membership, such as U3, the priority is monetization: since these users already spend considerable time in the app, targeted in-app purchase prompts, limited-time offers, or premium feature trials are likely to convert attention into revenue. For users with strongly dominant Purchase-cluster membership, such as U4, the priority is retention and re-engagement: since these users already demonstrate willingness to spend, notifications, content updates, or gamified challenges designed to increase visit frequency could substantially raise their lifetime value without needing to alter their purchase intent.

For boundary users with more balanced membership probabilities across both clusters, a blended strategy is appropriate — for instance, a moderate-value purchase prompt combined with light engagement nudges, since neither a pure engagement campaign nor a pure monetization campaign would be optimally matched to their mixed behavioural profile. Over time, the company can track how each user's membership vector shifts (for example, following a product update or a marketing campaign), using this movement as a direct, quantitative measure of whether an intervention successfully shifted behaviour in the desired direction — a level of measurement precision that a hard-clustering, single-label approach cannot provide.

Q4: Agricultural Crop Monitoring — Effect of Normalization on Clustering

1. Understanding of the Data

The dataset covers five farms (F1–F5) with three attributes measured on very different scales: Rainfall (mm, ranging 280–520), Soil Fertility Score (a small integer scale, ranging 3–9), and Crop Yield (tons, ranging 65–90). F1 (300, 8, 85) and F4 (280, 9, 90) show low rainfall but high fertility and high yield. F2 (500, 4, 70) and F5 (520, 3, 65) show high rainfall but low fertility and low yield. F3 (400, 6, 80) sits in between on all three attributes.

2. Clustering Behaviour Before Normalization

Distance-based clustering algorithms such as K-Means or hierarchical clustering compute similarity using metrics like Euclidean distance, which are highly sensitive to the numeric scale of each attribute. Here, Rainfall spans a range of roughly 240 units (280–520), while Soil Fertility spans only 6 units (3–9) and Crop Yield spans 25 units (65–90). Because the raw magnitude of Rainfall dwarfs the other two attributes, it dominates the distance calculation almost entirely — differences in fertility or yield contribute negligibly to the overall distance between farms compared to differences in rainfall.

As a result, clustering on unnormalized data effectively becomes clustering on Rainfall alone. Farms would be grouped primarily by how similar their rainfall values are: F1 and F3 might cluster together mainly because 300 and 400 are numerically closer than 300 is to 500, even though F3's fertility and yield profile is meaningfully different from F1's. Similarly, F2 and F5 would cluster tightly due to their close rainfall values (500 and 520), correctly in this case, but for the wrong (or at least incomplete) reason — the algorithm is not truly 'seeing' the fertility-yield relationship, only the rainfall proximity.

3. Clustering Behaviour After Normalization

Normalization (e.g., Min-Max scaling to a 0–1 range, or Z-score standardization) rescales each attribute so that all three contribute proportionally to the distance calculation, regardless of their original units or magnitude. After normalization, the true multi-attribute pattern in the data becomes visible:

- Cluster A (F1, F4): Low rainfall, high fertility, high yield — representing farms where good soil quality compensates for lower rainfall, achieving strong crop output.
- Cluster B (F2, F5): High rainfall, low fertility, low yield — representing farms where, despite abundant rainfall, poor soil fertility limits productivity.
- F3: An intermediate case, roughly equidistant from both clusters, reflecting its balanced (moderate) values across all three attributes.

This normalized clustering aligns with genuine agronomic insight — fertility appears to be a stronger driver of yield than rainfall alone in this dataset — a relationship that was masked before normalization because rainfall's larger numeric scale suppressed fertility's real influence on the clustering outcome.

4. Conclusion — Why Normalization Improves Reliability

Normalization does not change the underlying farms or their real-world characteristics; it changes how fairly each characteristic is represented in the distance metric that drives clustering. Without normalization, a variable with a large numeric range (Rainfall) can silently

dominate the algorithm's notion of similarity, producing clusters that reflect a single dominant variable rather than the true multi-dimensional structure of the data. This can mislead agricultural decision-makers into believing rainfall is the primary factor differentiating farms, when in fact soil fertility plays an equally or more important role.

By normalizing the data before clustering, the research center ensures that Rainfall, Soil Fertility, and Crop Yield all contribute proportionally to the analysis, producing clusters that genuinely reflect the combined agronomic profile of each farm. This leads to more reliable and actionable recommendations — for instance, targeting fertility-improvement programs at farms in Cluster B rather than assuming their lower yield is simply a rainfall problem — ultimately improving the quality of agricultural planning and resource allocation decisions.

5. Recommended Practice for Agricultural Data Analysis

This example generalizes to a broader best practice in agricultural (and indeed any multi-attribute) data analysis: whenever the variables being clustered are measured in different units or have substantially different numeric ranges, normalization should be treated as a mandatory preprocessing step rather than an optional refinement. Two common techniques are Min-Max scaling, which rescales every attribute to a fixed range such as 0–1, and Z-score standardization, which rescales each attribute to have a mean of 0 and a standard deviation of 1. Min-Max scaling is often preferred when the approximate upper and lower bounds of a variable are known and meaningful (as with rainfall or a bounded fertility score), while Z-score standardization is more robust when the data may contain outliers or when the true range of future values is uncertain.

For the agricultural research center, adopting normalization as a standard step before every clustering run would ensure that future analyses — potentially incorporating additional variables such as pesticide usage, irrigation type, or soil pH, each with its own very different scale — remain reliable and are not silently skewed by whichever variable happens to have the largest raw numeric range. Combined with periodic re-validation of cluster quality (for example, using silhouette scores as in Q5), this practice would give the center a robust, repeatable methodology for extracting genuine agronomic insight from increasingly rich farm-monitoring datasets.

Q5: Employee Performance Evaluation — Choosing K using Silhouette Scores

1. Understanding of the Data

The dataset contains five employees (E1–E5) evaluated on three metrics: Productivity Score, Attendance (%), and Skill Assessment Score. E5 (93, 96, 90) and E1 (88, 92, 84) show consistently high values across all three metrics, indicating top performers. E4 (55, 65, 58) shows consistently low values, indicating a struggling performer. E2 (62, 75, 68) is below average but not as low as E4, while E3 (78, 88, 72) sits at a solid, moderate performance level, closer to the high-performing group than to the low-performing one.

2. Analysis of the Silhouette Scores

The silhouette score measures how well each data point fits within its assigned cluster compared to neighbouring clusters, on a scale from -1 (poor fit) to +1 (excellent fit), with higher average scores indicating tighter, better-separated clusters. The scores provided are: K=2 → 0.48, K=3 → 0.68, K=4 → 0.52.

- K=2 (0.48): The lowest score of the three. With only two clusters, employees with genuinely different performance levels are likely being forced together — for example, the moderate performer E3 may be grouped with either the high or low performers despite not truly belonging to either extreme, reducing overall cohesion.
- K=3 (0.68): The highest score by a clear margin. Three clusters allow the natural three-tier structure in the data — High Performers, Moderate Performers, and Low Performers — to be captured accurately, producing tight, well-separated groups.
- K=4 (0.52): A drop from K=3. With only five employees, forcing four clusters means at least two clusters contain a single employee each, which can artificially inflate apparent separation for those singletons while simultaneously breaking apart genuinely similar employees (such as separating E1 and E5, or splitting a cohesive moderate group), reducing the overall average silhouette score and indicating over-fragmentation.

3. Application of Concepts — Determining the Most Suitable K

Since K=3 yields the highest silhouette score (0.68) — substantially higher than both K=2 (0.48) and K=4 (0.52) — it represents the most suitable number of clusters for this dataset. Applying K=3, the employees group naturally as follows:

- Cluster 1 – High Performers (E1, E5): Consistently high productivity, attendance, and skill scores (84–93 range across metrics).
- Cluster 2 – Moderate Performers (E3): Solid, above-average scores (72–88 range) that are distinct enough from both extremes to warrant their own category.
- Cluster 3 – Low Performers (E2, E4): Below-average scores across all three metrics (55–75 range), indicating a clear need for support or development.

4. Conclusion and Justification

The choice of K=3 is justified both statistically and practically. Statistically, it produces the highest silhouette score among the tested values, meaning employees are, on average, much more similar to others within their own cluster than to those in neighbouring clusters — a direct, quantitative indicator of cluster quality. Practically, three clusters map naturally onto a performance-management framework that organizations commonly use: top performers (candidates for recognition, promotion, or retention initiatives), moderate performers

(candidates for targeted skill development to help them progress into the top tier), and low performers (candidates for structured performance-improvement plans or additional training and mentorship).

Choosing $K=2$ would blur the important distinction between moderate and low performers, potentially causing an organization to under-invest in employees like E3 who are close to becoming top performers, or to over-generalize support strategies. Choosing $K=4$ would over-segment a small dataset, creating clusters that are not statistically robust and that may lead to inconsistent or overly granular HR policies. By selecting $K=3$, supported by the highest silhouette score, the company obtains a clustering model that is both statistically sound and directly actionable for designing differentiated, effective employee-performance-evaluation and development strategies.

5. Contribution of the Selected Model to Effective Evaluation

Beyond simply grouping employees, the $K=3$ model contributes to effective performance evaluation by providing an objective, data-driven foundation that complements — rather than replaces — managerial judgement. Because the clustering is based on three independent, quantifiable metrics (Productivity, Attendance, and Skill Assessment) rather than a single score, it reduces the risk of evaluation bias that can arise from focusing on only one dimension of performance, such as productivity alone, which might unfairly penalize an employee with excellent attendance and skill but a temporarily lower output due to external factors. The silhouette score further gives management a transparent, defensible justification for why employees were grouped as they were, which is valuable when communicating evaluation outcomes or designing development plans that need to be perceived as fair and consistent.

Operationally, the three resulting clusters can feed directly into differentiated HR actions: High Performers can be fast-tracked for leadership development or retention bonuses to reduce attrition risk; the Moderate Performer can be given a focused, achievable improvement plan targeting the specific metric(s) holding them back from the top cluster; and Low Performers can receive structured mentorship, additional training, or closer supervision, with progress re-measured in the next evaluation cycle to see whether their metrics — and consequently their cluster membership — improve over time. In this way, silhouette-guided cluster selection transforms a routine statistical exercise into a repeatable, evidence-based performance-management process.